CSE2023-HW4

(1) $P(n): 2^n > n^2$ for $n>4$ $\quad P(5): 2^5 = 32 > 5^2 = 25$ TRUE

k is an arbitrary integer

$P(k+1): 2^{k+1} > (k+1)^2 \quad k+1>4,\ n>4 \rightarrow k>4$

$2 \cdot 2^k > k^2+2k+1$ $\quad k>4$ $\quad k>4 \rightarrow 2k>1$

$k^2 > 4k$

$k^2 > 2k+2k$

$k^2 > 2k+1$

(2)

a) $24, \ldots, 120$

$a_n = 21+3n \quad 1 \le n \le k$

$22 < 21+3n < 121$

$a_1 = 24$

$a_{33} = 120$

$120 = 21+3n$

$99 = 3n$

$n = 33 = k$

(33)

b) $24, \ldots, 120 \quad a_n = 20+4n \quad 1 \le n \le k$

$22 < 20+4n < 121 \quad a_1 = 24$

$120 = 20+4n$

$100 = 4n$

$n = 25 = k$

(25)

(3)

6 ones, 8 zeros: $C(14,6) = 3003$

7 ones, 7 zeros: $C(14,7) = 3432$

8 ones, 6 zeros: $C(14,8) = 3003$

9 ones, 5 zeros: $C(14,9) = 2002$

total = **11440**

(4) $(80 \times 40) + 1 =$ **3201**

(5)

a) $\binom{10}{3} \cdot 8^3 \cdot 21^7 = 110658879959040$

b) $\binom{10}{4} \cdot 21^4 \cdot 8^6 = 107062257254 40$

c) $29^{10} - \left(21^{10} + \binom{10}{1} \cdot 8^1 \cdot 21^9\right) = 340\,484\,948\,595\,520$

d) $29^{10} - \left(21^{10} + \binom{10}{1} \cdot 8^1 \cdot 21^9 + \binom{10}{2} \cdot 8^2 \cdot 21^8\right) = 231\,555\,113\,635\,840$

e) $29^{10} - \left(8^{10} + \binom{10}{1} \cdot 21^1 \cdot 8^9 + \binom{10}{2} \cdot 21^2 \cdot 8^8\right) = 420345029983977$